

Parallelization of Diophantine Solvers

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May 5, 2026

Introduction

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Answer: Yes! The smallest was found in 2002:

$$1117^6 + 770^6 = 1092^6 + 861^6 + 602^6 + 212^6 + 84^6$$

Resta-Meyrignac Algorithm

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Finally, we define $v = a_1^6 + a_2^6 - (b_1^6 + b_2^6)$ and check if we can decompose $v/7^6 = c_1^6 + c_2^6 + c_3^6$.

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The smallest solutions to the diophantine equation

$$x^6 + y^6 = a^6 + b^6 + c^6 + d^6 + e^6.$$

Math. Comput., 72(242):1051–1054, Apr. 2003.